

Prep Test 2: Functions and Polynomial Graphs

75 minutes | 100 points | 10 questions

Name: _____

Date: _____

Directions

- Show your work.
- Use interval notation where requested.
- Questions 6 and 8 require qualitative polynomial sketches. Exact extrema and a uniform vertical scale are not required.
- Suggested question times total 70 minutes, with a final 5 minutes for review.

Use the space provided; additional work may go on the back.

Suggested timing

Question	Points	Minutes
1. Graph a piecewise function	10	6
2. Transform a given graph	12	6
3. Graph both composition orders	14	9
4. Evaluate and analyze two graphs	10	7
5. Compare two algebraic compositions	6	5
6. Construct and sketch a polynomial	14	10
7. Compose functions and find a domain	8	6
8. Analyze and sketch a quartic	8	6
9. Use a composite model to meet a budget	10	9
10. Construct and graph a square-root function	8	6

Final review: 5 minutes. Total: 75 minutes.

Question 1

6 minutes | 10 points

1

Graph a piecewise function

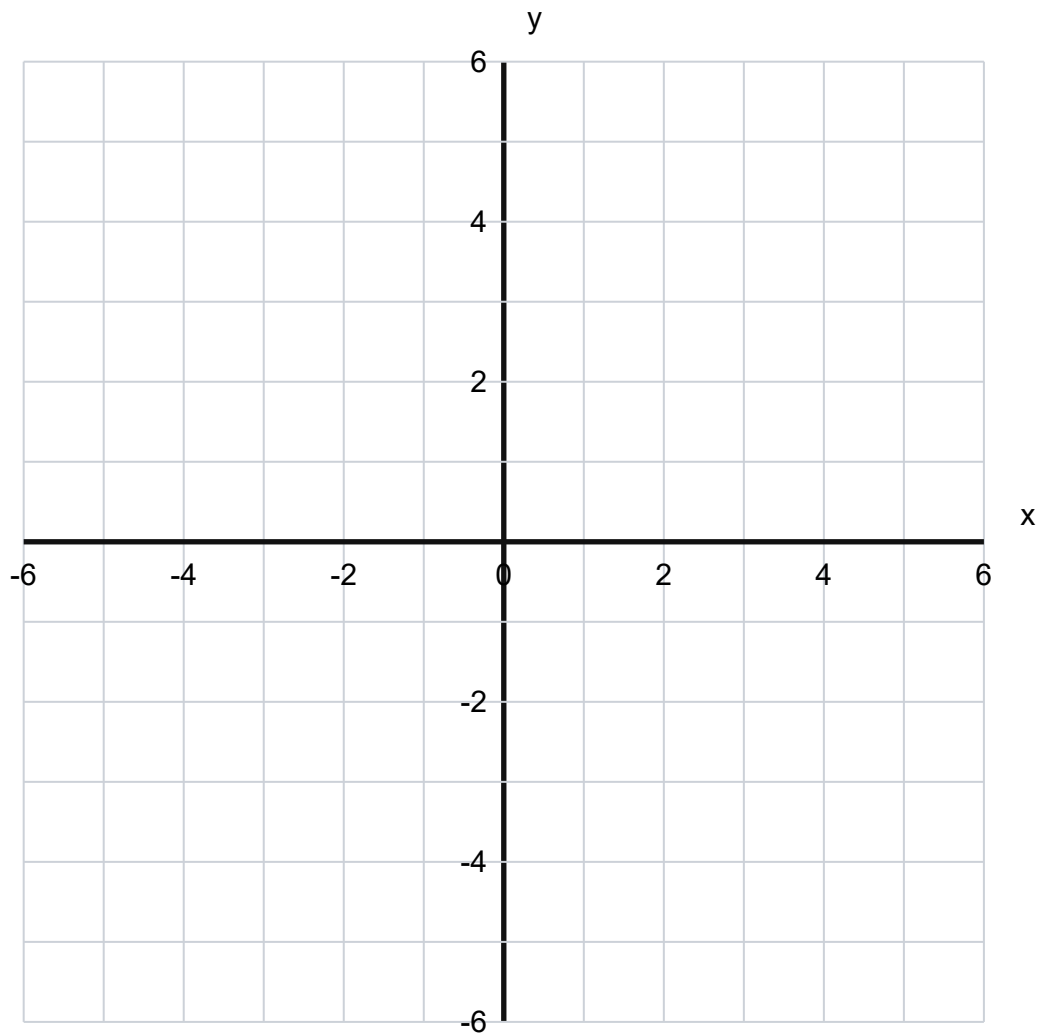
10 points

Let $p(x)=2x+4$ for $x<-1$, and $p(x)=3$ for $x\geq-1$.

- (a) Find $p(-2)$, $p(-1)$, and $p(2)$. Show which branch you use.
- (b) Graph p on axes $x=-6$ to 6 , $y=-6$ to 6 , with one unit per square. Continue each branch with an arrow. Label the boundary coordinates and show open or closed circles.
- (c) Give the domain and range in interval or set notation.

Question 1: your graph

One unit per square.



Answers and work

Question 2

6 minutes | 12 points

2

Transform a given graph

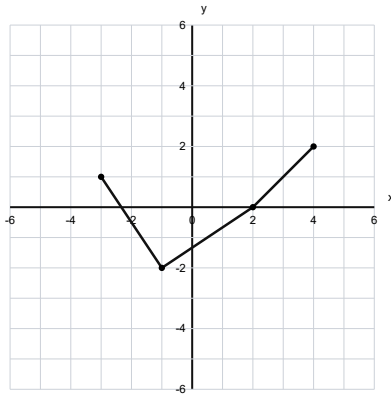
12 points

The graph of f is shown below. Its vertices are $(-3,1)$, $(-1,-2)$, $(2,0)$, and $(4,2)$, joined by straight segments, with included endpoints. Let $h(x) = -f(x - 2) + 1$.

(a) Graph h on the provided grid. Label all vertices with their coordinates and mark the endpoints. Use axes $x=-8$ to 8 , $y=-8$ to 8 , one unit per square.

(b) Give the domain and range of h .

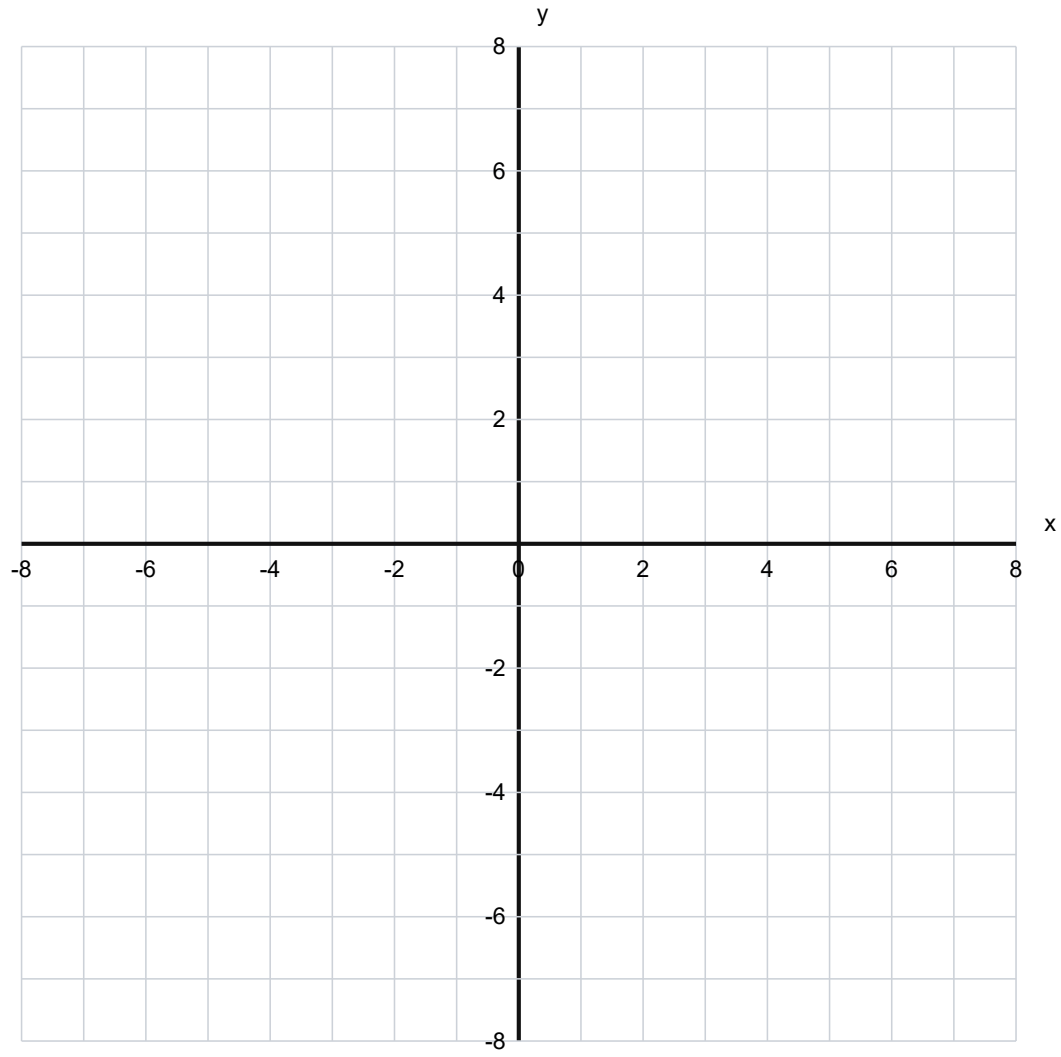
Given graph: f



Work space

Question 2: your graph

One unit per square.



Answers and work

Question 3

9 minutes | 14 points

3 Graph both composition orders

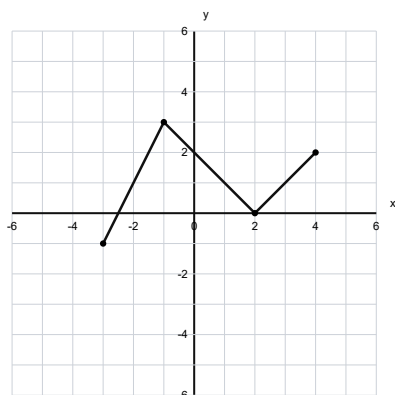
14 points

The graph of f consists of straight segments through $(-3,-1)$, $(-1,3)$, $(2,0)$, $(4,2)$, in that order, with included endpoints. Let $g(x)=1-2x$.

(a) Graph $f(g(x))$ and $g(f(x))$ on the provided grid. Use a solid line for $f(g(x))$ and a dashed line for $g(f(x))$. Label the vertices and both graphs. The axes run from -6 to 6 , one unit per square.

(b) Give both domains and describe the transformations of f in each composition. Show your work.

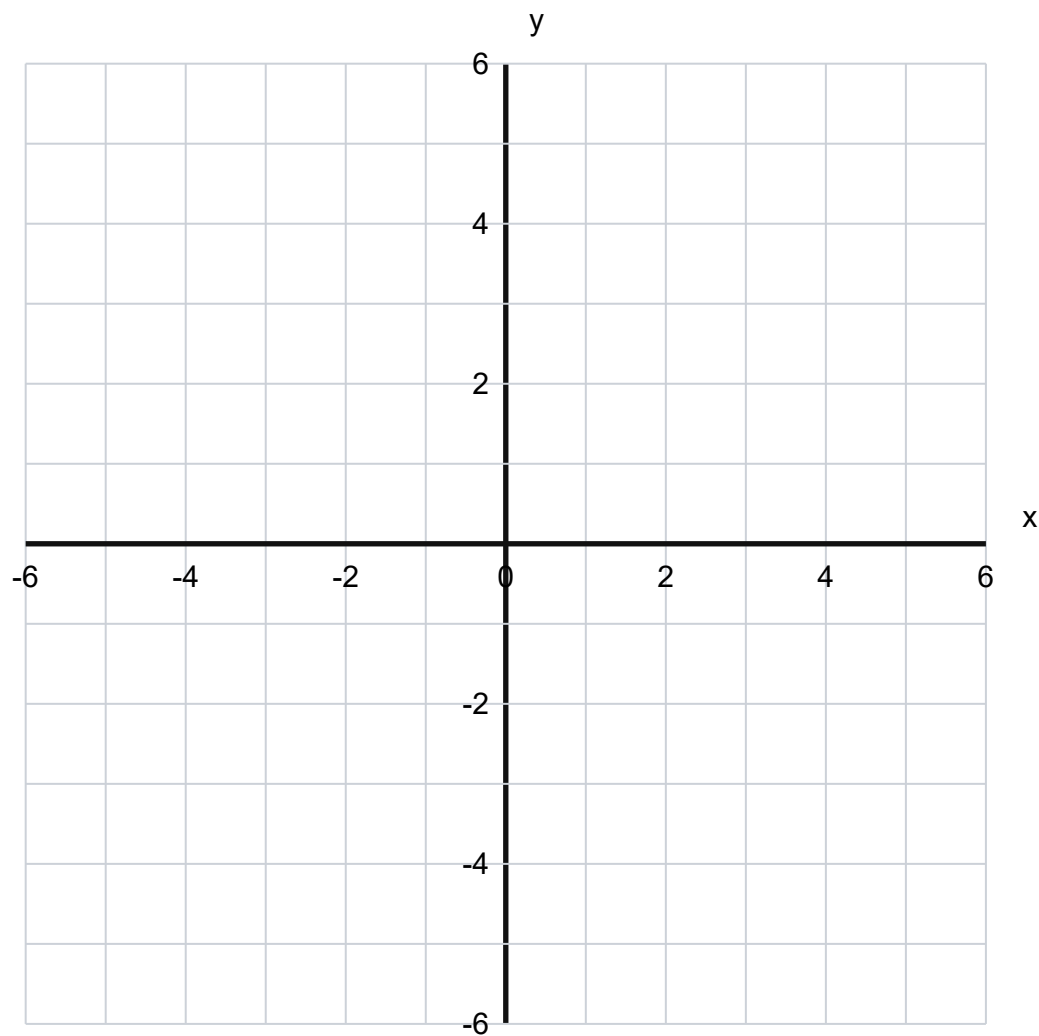
Given graph: f



Work space

Question 3: your graph

One unit per square.



Answers and work

Question 4

7 minutes | 10 points

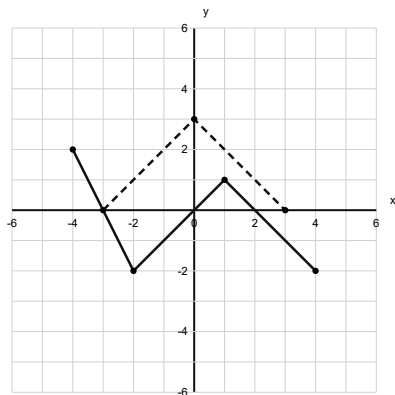
4 Evaluate and analyze two graphs

10 points

Both graphs consist of straight segments joining the listed vertices, with included endpoints. f : $(-4,2)$, $(-2,-2)$, $(1,1)$, $(4,-2)$. g : $(-3,0)$, $(0,3)$, $(3,0)$. The graphs below use axes $x=-6$ to 6 , $y=-6$ to 6 , one unit per grid line.

- Find $g(f(-3.5))$ and $f(g(-3.5))$. Show your work.
- Find all x -values for which $g(f(x))=2$. Show your work.
- Give all zeros of f and its increasing and decreasing intervals.

Given graph: f solid; g dashed



Work space

Answers and work

Question 5

5 minutes | 6 points

5

Compare two algebraic compositions

6 points

Let $A(x) = x^2 - 2x$ and $B(x) = 2x - 1$.

(a) Find and simplify $A(B(x))$ and $B(A(x))$.

(b) Find all real x -values for which $A(B(x)) = B(A(x))$. Justify your answer.

Answers and work

Question 6

10 minutes | 14 points

6

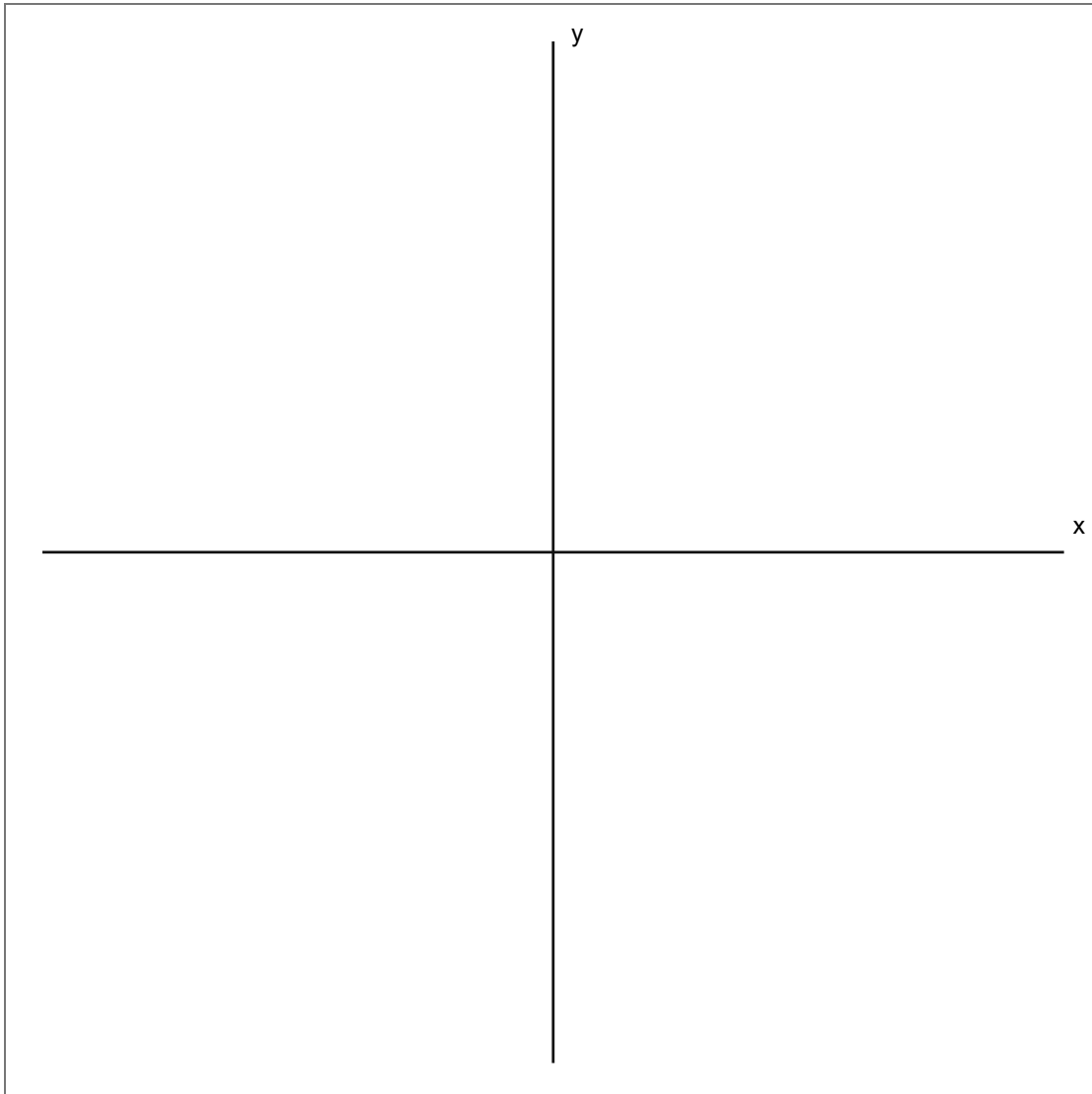
Construct and sketch a polynomial

14 points

A degree-5 polynomial P has exactly two distinct real zeros: -2 with multiplicity 2, and 1 with multiplicity 3. It passes through $(0, -8)$.

- (a) Find P in factored form and show how you determine its leading factor.
- (b) Make a qualitative sketch. Label both zeros and the y -intercept, show the correct ends, and show the graph's behavior at each zero. Exact extrema are not required; the sketch need not use a uniform vertical scale.
- (c) State the intervals where $P(x) > 0$ and $P(x) < 0$. Explain where the sign changes and where it does not.

Question 6: your qualitative sketch



Answers and work

Question 7

6 minutes | 8 points

7

Compose functions and find a domain

8 points

Let $f(t) = \sqrt{t}$, $g(x) = 9 - x^2$, and $H(x) = \frac{f(g(x))}{g(x)-5}$.

- (a) Write H as one expression in x.
- (b) Find the domain in interval notation. Show why each excluded value is excluded.
- (c) Is x=3 allowed? Is x=2 allowed? Explain using the original expression.

Answers and work

Question 8

6 minutes | 8 points

8

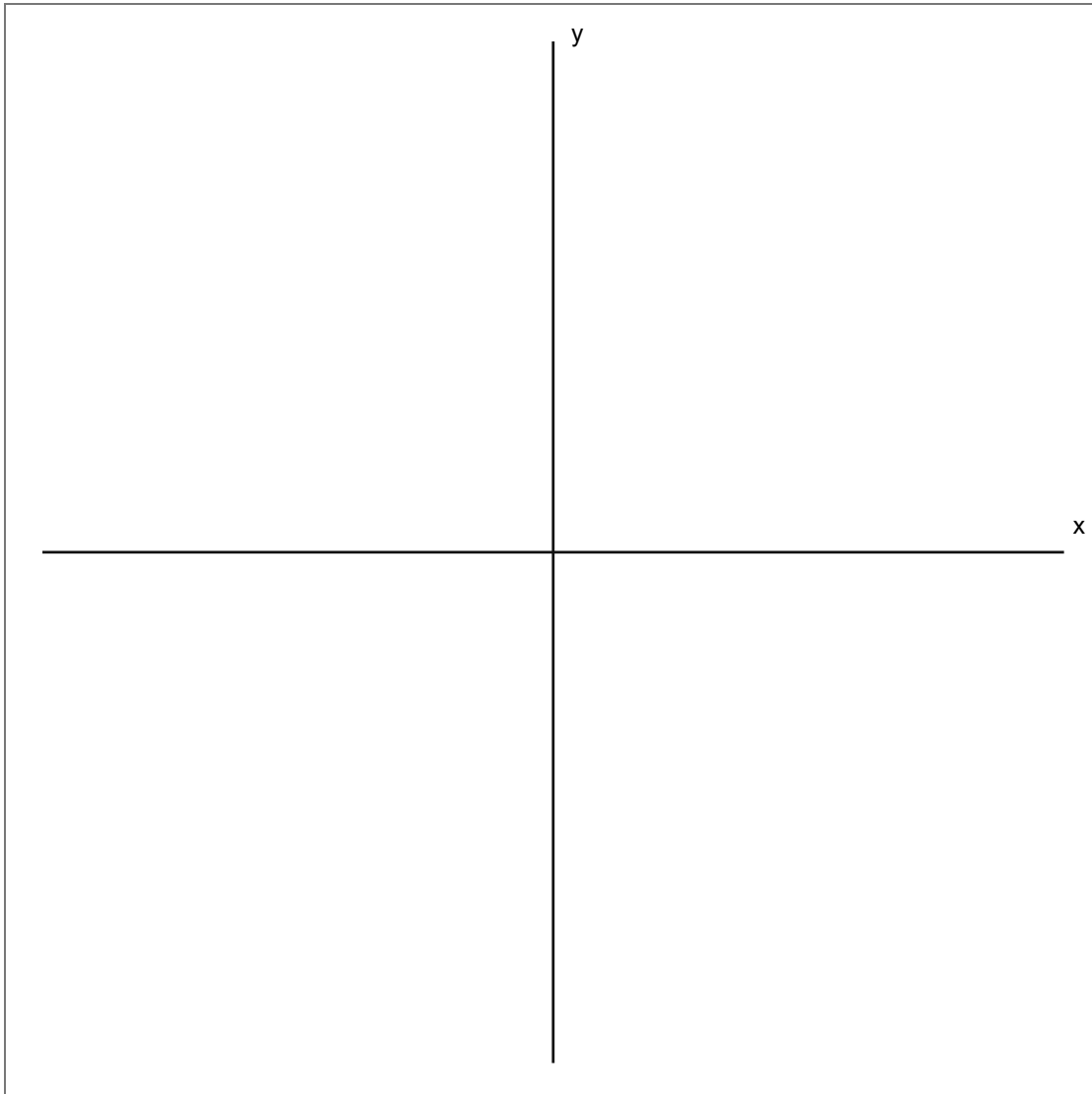
Analyze and sketch a quartic

8 points

Let $R(x) = -(x + 2)(x - 1)^2(x - 3)$.

- (a) State the degree, leading coefficient, zeros, and multiplicities. Find the y-intercept.
- (b) Make a qualitative sketch with all intercepts labeled and arrows on both ends. Show which zeros cross and which touch. Exact extrema and a uniform vertical scale are not required.
- (c) State the intervals where $R(x) > 0$ and $R(x) < 0$. Compare its end behavior with the polynomial in Question 6.

Question 8: your qualitative sketch



Answers and work

Question 9

9 minutes | 10 points

9

Use a composite model to meet a budget

10 points

A rectangular planting bed has perimeter 24 feet. Its width is w feet, with $2 \leq w \leq 6$, and its length is $12-w$ feet. Its area is $A(w) = w(12-w)$ square feet.

Soil delivery costs $C(a) = 8a$ dollars when $0 \leq a \leq 30$ square feet, and $C(a) = 240 + 5(a-30)$ dollars when $a > 30$ square feet. There are no other charges.

- (a) Write the composition giving cost in terms of width. Explain the order using units.
- (b) Write that cost as a piecewise function of w , including the width intervals for each price rule. State the model domain.
- (c) Find the cost when $w=4$.
- (d) With a budget of \$260, find the greatest permitted width. Give an exact value and a decimal to the nearest hundredth. Justify the branch and domain used.

Answers and work

Question 10

6 minutes | 8 points

10 Construct and graph a square-root function

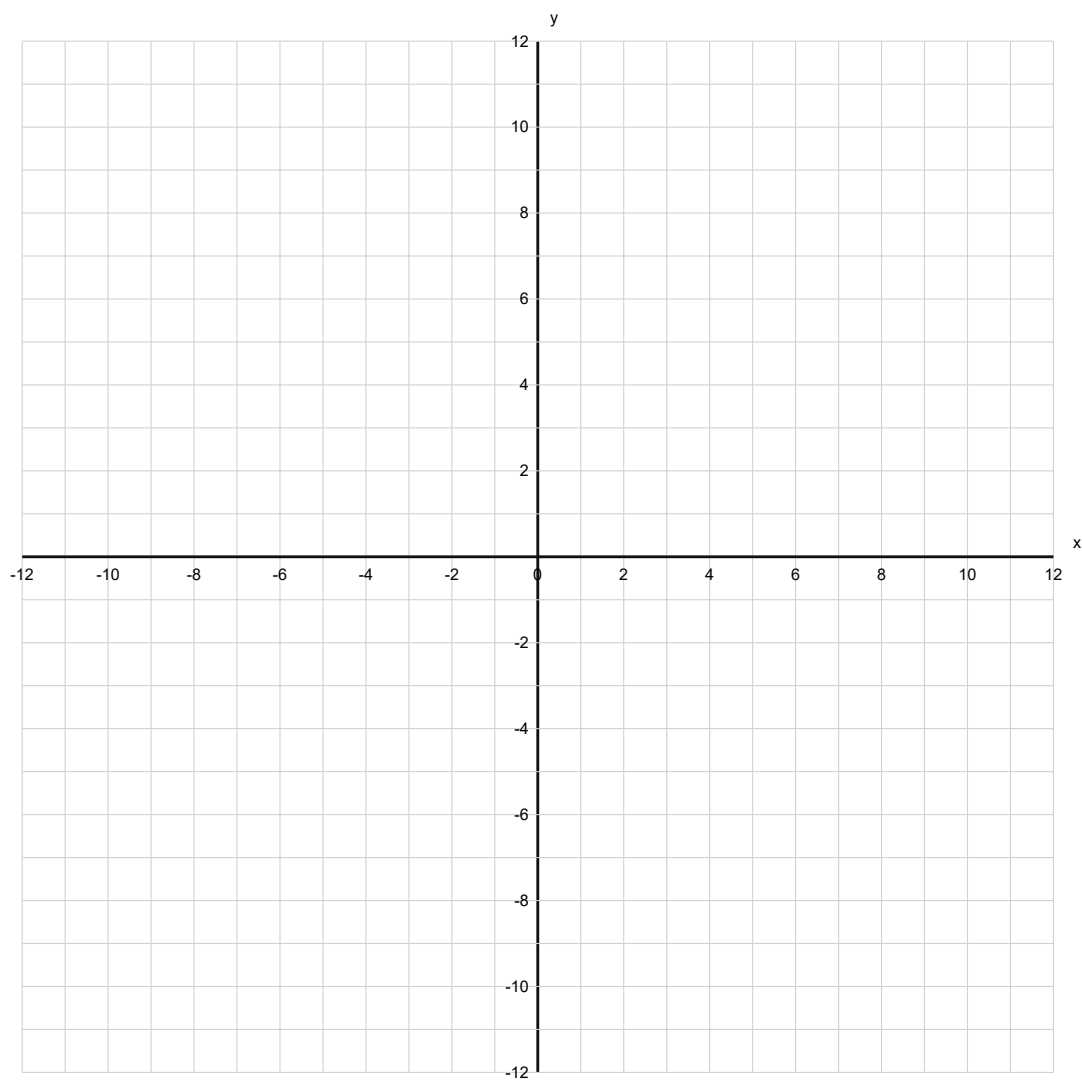
8 points

A square-root function has the form $r(x) = a\sqrt{x - h} + k$, domain $[1, \infty)$, range $(-\infty, 3]$, and passes through $(5, -1)$.

- (a) Find its equation. Show your work.
- (b) Graph r on the provided grid, with both axes from -12 to 12 and one unit per square. Label its starting point and three other points.
- (c) State its x-intercept exactly and its interval of decrease.

Question 10: your graph

One unit per square.



Answers and work
